
MEASUREMENT PROTOTYPE

Geographic Household Pressure Index

A State-Level Public-Data Measurement Prototype

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Abstract

This note revises a first geographic household-resilience map into a narrower measurement prototype: a state-level **Geographic Household Pressure Index**. The change in title is substantive. The current public-data object does not directly measure household resilience, which would require evidence on households' ability to absorb income, expense, housing, health, or labor-market shocks without severe welfare loss. It instead measures selected state-level pressures that plausibly condition resilience: renter rent burden, an FHFA house-price-index-to-income proxy, median household income rank, under-35 homeownership, and under-35 household formation. The index is transparent and useful as a diagnostic layer, but it is not causal, not a welfare ranking, not a recommendation about where to live, and not yet a validated resilience measure. Robustness checks show that the headline index is stable to excluding the FHFA proxy and to a housing-heavy weighting scheme, but not reducible to any single component. Simple panel validation exercises using 2011 pressure to explain 2011–2024 changes are mixed and should be treated as a warning rather than a success claim. The project is therefore best understood as an applied measurement build in progress: promising enough to continue, but not publishable as a finished economics paper until stronger validation, uncertainty, local geography, and outcome links are added.

1. Motivation and revised contribution

Geographic conditions shape household financial exposure. A national household with the same nominal income, savings habit, and debt posture can face different practical pressure depending on local rents, home-entry costs, wage levels, labor markets, transportation demands, insurance costs, and regional price levels. National averages therefore miss an important feature of household vulnerability: pressure is place-based.

The original proof of concept asked whether public state-level data could produce a credible first-pass map of household resilience pressure. The answer remains yes, but the external review correctly identified a more important distinction: the current object should not be called a resilience map. Resilience is an outcome or capacity. It concerns whether households can maintain ordinary obligations and absorb shocks without quickly falling into missed payments, forced moves, high-cost borrowing, food insecurity, delayed care, or loss of housing and mobility. The current index does not observe those outcomes directly.

This revised note therefore makes a narrower claim. It constructs a **Geographic Household Pressure Index**: a transparent state-level composite of public proxies that may condition household resilience. The contribution is not an journal-level causal finding. It is an auditable measurement

prototype that tests whether a geographic pressure layer can be built, decomposed, and stress-tested before stronger data are added.

The revision responds to the core review concerns in four ways. First, it renames the object from resilience to pressure. Second, it treats the FHFA HPI-income variable as a potentially imperfect appreciation-pressure proxy rather than a direct house-price-level affordability measure. Third, it adds robustness checks across alternative specifications. Fourth, it adds a simple panel validation exercise and reports the mixed results honestly.

2. Conceptual object: pressure, vulnerability, and resilience

This note distinguishes three concepts.

Pressure refers to observable conditions that make ordinary household obligations harder to maintain: high rent burden, high housing-price pressure relative to income, weak income position, or delayed household formation among young adults.

Vulnerability is the household-level risk that pressure becomes hardship when a shock occurs. Vulnerability depends on liquid buffers, debt-service burdens, family structure, insurance, health, labor-market attachment, and informal support.

Resilience is the realized or potential capacity to absorb shocks without severe welfare loss. It must ultimately be validated against outcomes such as emergency-savings shortfalls, arrears, delinquencies, eviction risk, food insecurity, delayed care, forced moves, or household-formation delay.

The present index measures pressure, not vulnerability or resilience. The practical motivation is that pressure is easier to observe geographically using public data. The empirical burden for future work is to show whether these pressure proxies predict independent vulnerability or resilience outcomes better than simpler measures.

This distinction is especially important for the under-35 variables. Low under-35 homeownership can reflect entry-cost barriers, but it can also reflect urban labor-market sorting, schooling, delayed marriage, immigration, rental amenities, mobility preferences, or demographic composition. Low under-35 household formation can reflect financial strain, but it can also reflect age structure or local preferences. These variables remain in the prototype because they capture a plausible household-entry channel, but the revised interpretation is cautious: they are pressure proxies, not direct welfare measures.

3. Data and construction

The prototype uses the existing state-year ACS/FHFA panel assembled for the public-data household research project. The source panel includes ACS state household and housing variables and FHFA state house-price-index data merged into a state-year file. The revised prototype uses the 2024 state/DC cross-section for the map and the 2011–2024 panel for simple validation checks.

The headline index uses two component groups.

Housing pressure combines:

- the percentile rank of renter rent-burden pressure, measured as the share of renter households with gross rent at or above 30 percent of income;
- the percentile rank of the FHFA HPI-income proxy.

The housing component is:

$$\text{Housing pressure} = 0.60 \times \text{rent-burden percentile} + 0.40 \times \text{HPI-income-proxy percentile}.$$

The FHFA term should be read carefully. It is not a direct median home-price level or mortgage-payment affordability measure. It is an index-based pressure proxy. Because this is a real measurement concern, the revised robustness table includes a specification that excludes the FHFA proxy entirely.

Income and formation pressure combines:

- reverse percentile rank of median household income;
- reverse percentile rank of under-35 homeownership;
- reverse percentile rank of under-35 household-formation share.

The income/formation component is the simple average of those three ranks.

The headline index is:

$$\text{Core pressure index} = 0.50 \times \text{housing pressure} + 0.50 \times \text{income/formation pressure}.$$

Higher values indicate higher measured pressure in this limited proxy set. Bands are fixed judgment thresholds for prototype communication, not statistically estimated categories: below 20 = lowest measured pressure; 20 to below 35 = lower; 35 to below 55 = middle; 55 to below 75 = elevated; 75 and above = highest measured pressure. Tables report whole-number interpretation in the prose even where decimals appear for auditability.

All ranks are unweighted state/DC percentile ranks. That choice makes the state-level diagnostic easy to audit, but it means the current output does not represent the number of households exposed nationally. A population-weighted version is a necessary next robustness step.

4. Prototype map and decomposition

Prototype geographic household-pressure index, 2024

Higher values combine rent burden, HPI-to-income pressure, lower income rank, and young-household formation stress. Tile map is schematic.

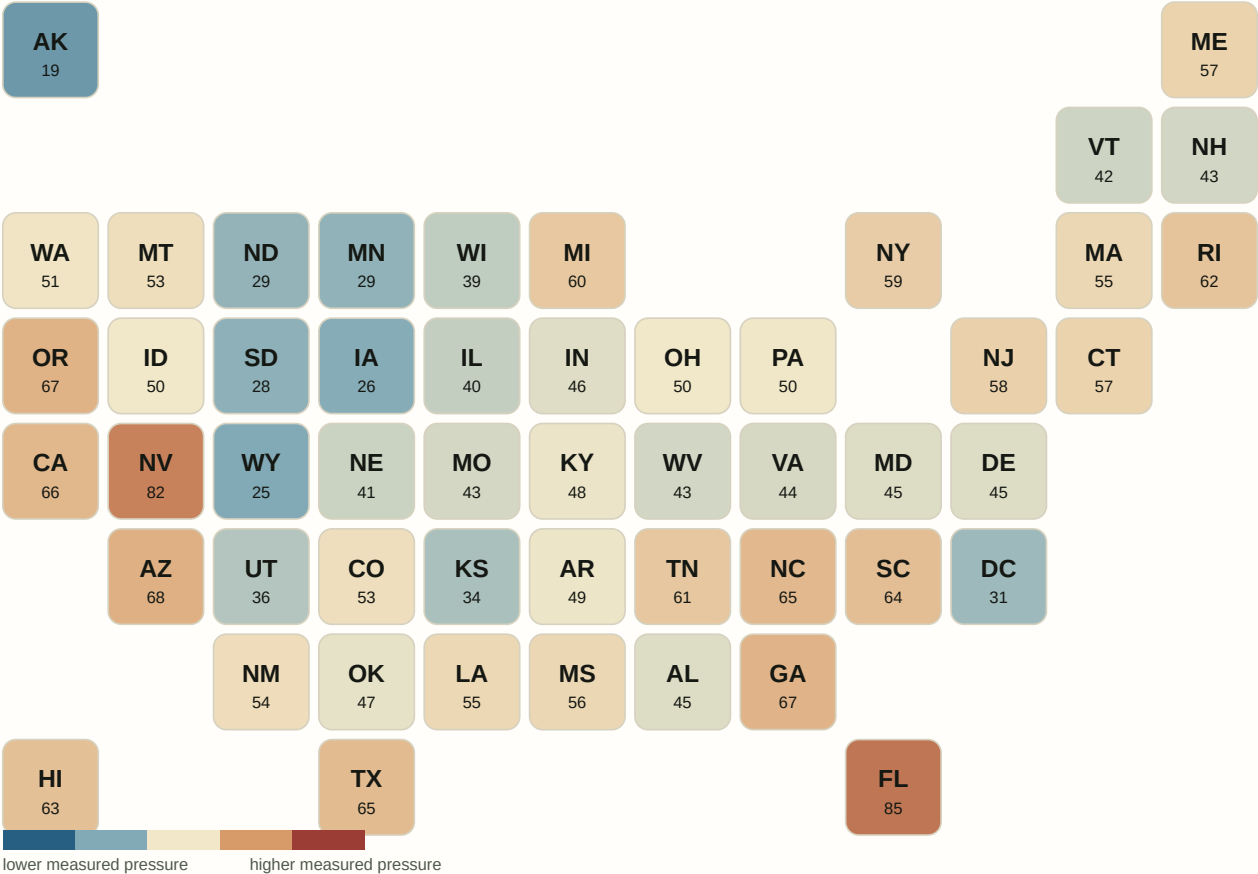


Figure 1. Prototype state household-pressure tile map, 2024. Higher values combine rent burden, HPI-income pressure, lower income rank, under-35 homeownership pressure, and under-35 household-formation pressure. The tile geography is schematic and should not be read as a precise geographic choropleth.

Figure 1 shows the state/DC pressure index as a schematic tile map. The figure is useful for diagnosis, but it should not be used as a leaderboard. State averages are coarse. Large states contain multiple housing markets, labor markets, and commuting zones; state-level scores can hide severe within-state exposure among renters, low-income households, and specific metro areas.

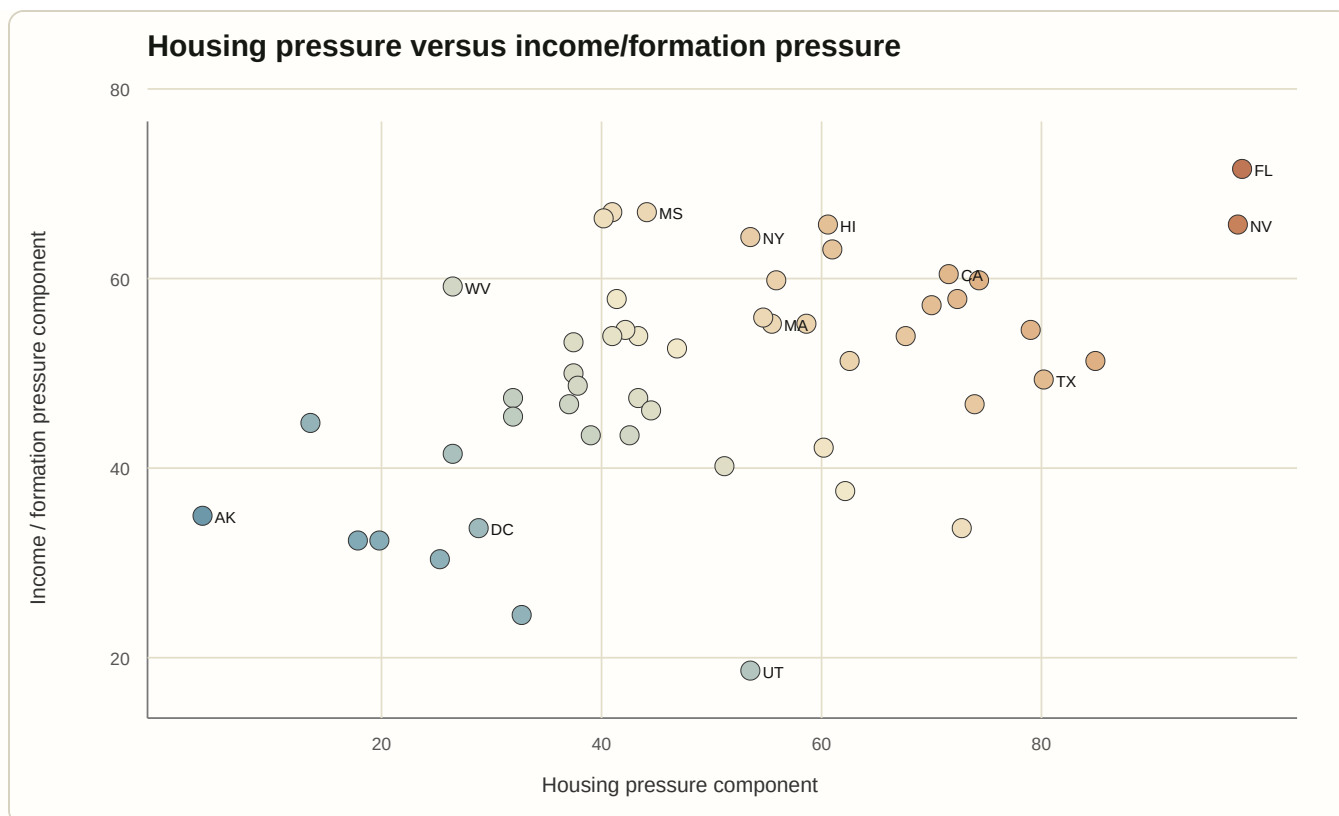


Figure 2. Housing pressure versus income/formation pressure. Points are states/DC. Labels highlight high-pressure, low-pressure, and selected large states.

Figure 2 is more informative than the map alone because it separates housing pressure from income/formation pressure. A state can appear elevated because housing pressure is high, because income/formation pressure is high, or because both channels stack together. This decomposition should remain central in any public version.

Table 1. Highest measured-pressure states in the proof-of-concept index, 2024.

State	Index	Band	Housing	Income/form.
Florida	84.90	highest measured pressure	98.24	71.57
Nevada	81.76	highest measured pressure	97.84	65.69
Arizona	68.10	elevated measured pressure	84.90	51.31
Oregon	67.06	elevated measured pressure	74.31	59.80
Georgia	66.80	elevated measured pressure	79.02	54.58
California	66.01	elevated measured pressure	71.57	60.46
North Carolina	65.10	elevated measured pressure	72.35	57.84
Texas	64.77	elevated measured pressure	80.20	49.35
South Carolina	63.59	elevated measured pressure	70.00	57.19
Hawaii	63.14	elevated measured pressure	60.59	65.69

Table 2. Lowest measured-pressure states in the proof-of-concept index, 2024.

State	Index	Band	Housing	Income/form.
Alaska	19.35	lowest measured pressure	3.73	34.97
Wyoming	25.10	lower measured pressure	17.84	32.35
Iowa	26.08	lower measured pressure	19.80	32.35
South Dakota	27.84	lower measured pressure	25.29	30.39
Minnesota	28.63	lower measured pressure	32.75	24.51
North Dakota	29.15	lower measured pressure	13.53	44.77
District of Columbia	31.24	lower measured pressure	28.82	33.66
Kansas	33.99	lower measured pressure	26.47	41.50
Utah	36.08	middle measured pressure	53.53	18.63
Wisconsin	38.69	middle measured pressure	31.96	45.42

The top and bottom tables are diagnostic checks, not welfare rankings. The low score for the District of Columbia in the earlier draft is a useful warning: high nominal income and unusual young-household composition can make the prototype understate pressure in high-cost urban jurisdictions. That is precisely why regional price parity, actual housing price levels, and within-state heterogeneity are not optional in the next build.

5. Robustness checks

The external review correctly noted that one housing-heavy sensitivity check was insufficient. Table 3 expands the robustness exercise by comparing the headline index with six alternative specifications: excluding the FHFA proxy, housing-heavy weighting, and single-component benchmarks.

Table 3. Specification robustness, 2024. Correlations compare each alternative with the headline equal-weight index. “Same band states” reports how many of 51 jurisdictions remain in the same broad band.

Specification	Pearson	Spearman	Same band states	N
rent income formation no hpi	0.91	0.90	42	51
housing heavy	0.99	0.99	49	51
rent burden only	0.77	0.77	16	51
income only	-0.00	-0.02	8	51
formation only	0.66	0.62	18	51
hpi income proxy only	0.50	0.47	16	51

Two results matter. First, excluding the FHFA HPI-income proxy leaves the headline index highly correlated with the no-HPI alternative. This reduces, but does not eliminate, concern that the FHFA proxy is driving the map. Second, the headline index is not equivalent to rent burden alone, income alone, formation alone, or the HPI proxy alone. That is reassuring for nonredundancy, but it is not validation. A composite can be nonredundant and still not predict meaningful outcomes.

The housing-heavy check remains useful as a first-order stress test. It shows that weighting choices affect some states, but the broad headline classification is stable for most jurisdictions.

Table 4. Largest sensitivity differences under a housing-heavy specification, 2024.

State	Equal-weight	Housing-heavy	Difference
Colorado	53.20	59.07	5.86
Utah	36.08	41.31	5.24
Arizona	68.10	73.14	5.04
West Virginia	42.81	37.91	-4.90
Nevada	81.76	86.59	4.82
North Dakota	29.15	24.46	-4.69
Alaska	19.35	14.66	-4.69
Texas	64.77	69.40	4.63
Michigan	60.33	64.41	4.08
Florida	84.90	88.90	4.00
Montana	53.27	49.35	-3.92
New Mexico	53.99	50.08	-3.90

The right interpretation is modest. The index is reasonably stable to a few transparent alternatives, but a publishable measurement paper would need a full specification grid, leave-one-variable-out table, z-score versus percentile-rank comparison, population-weighted version, and band-switch uncertainty analysis.

6. First panel validation check

The source data are a panel, so the revised draft adds a simple validation exercise. The test asks whether a 2011 pressure measure predicts 2011–2024 changes in related outcomes. To reduce mechanical overlap, the 2011 pressure measure used here averages rent-burden pressure, income pressure, and HPI-income-proxy pressure; it excludes under-35 formation variables when predicting changes in under-35 outcomes.

Table 5. Simple panel validation checks. Correlations compare 2011 pressure with 2011–2024 state-level changes.

Outcome	Pearson	Spearman	N
Δ under-35 homeownership, 2011–2024	0.37	0.40	51
Δ under-35 household-formation share, 2011–2024	0.12	0.02	51
Δ rent-burden rate, 2011–2024	-0.19	-0.18	51
Δ HPI-income proxy, 2011–2024	0.29	0.25	51

These checks are intentionally preliminary and the results are mixed. The correlations do not establish that the index measures resilience. They also do not show strong predictive power for household formation changes. The exercise is nevertheless useful because it disciplines the project: if future versions cannot predict independent hardship, liquidity, housing-instability, or formation outcomes better than simpler benchmarks, the composite index should not be presented as a validated measure.

A stronger validation design should compare rent burden alone, median income alone, direct housing-price-to-income measures, and the full composite against independent outcomes such as emergency-savings shortfalls, rent arrears, eviction filings, mortgage and credit-card delinquencies, bankruptcy, food insecurity, utility shutoffs, homelessness risk, disaster displacement, labor-force exits, household formation delays, and migration responses. State-level validation may remain imperfect; metro, county, commuting-zone, or microdata validation will likely be required.

7. Measurement limitations

The current prototype has five material limitations.

First, the geographic unit is too coarse for many economic questions. States are administrative units, not housing markets or labor markets. A state-level map can be useful for a first dashboard, but serious analysis should move toward metropolitan areas, commuting zones, counties, or Public Use Microdata Areas while modeling the tradeoff between granularity and measurement error.

Second, the FHFA HPI-income proxy is not a direct price-level affordability measure. Future versions should replace or supplement it with actual median home value, median sale price, owner-cost burden, mortgage-payment-to-income arithmetic, or properly normalized price-to-income ratios.

Third, the prototype lacks statistical uncertainty. ACS estimates have sampling error, and rank/band assignments should eventually incorporate margins of error or bootstrap uncertainty. The correct question is not whether one state ranks above another by a few points, but whether broad pressure classifications are stable under plausible sampling and specification uncertainty.

Fourth, the index lacks important pressure channels: labor-market stress, commute burden, regional price parity, insurance and disaster exposure, debt service, and liquid buffers.

Fifth, the literature grounding is still preliminary. A finished paper should engage more deeply with household finance research on liquidity and hand-to-mouth behavior; urban economics research on housing affordability, local labor markets, and spatial equilibrium; work on regional inequality and cost-of-living measurement; and hardship-measurement research using surveys and administrative outcomes.

8. Intended use

The intended use is not to tell households where to live or to rank states. The intended use is a diagnostic context layer for researchers and planners who need to know whether household pressure is geographically concentrated and which components drive it.

Different use cases would require different versions. Housing policy would need a stronger rent, owner-cost, supply, and local-wage design. Emergency planning would need disaster and insurance exposure. Macroeconomic resilience work would need debt service, liquid buffers, and interest-rate exposure. A household-facing planner would need local costs and household-specific inputs rather than state averages.

This prototype is therefore best viewed as infrastructure: a transparent starting point for a future validated geographic pressure measure.

9. Next empirical build

The next version should add four improvements before publication as a public map or paper.

1. **Better housing levels.** Replace or supplement the FHFA HPI-income proxy with median home value, median sale price, owner-cost burden, mortgage-payment-to-income estimates, and rent-to-income measures.
2. **Labor-market, commute, and regional-price context.** Add BLS LAUS unemployment, ACS commute variables, and BEA regional price parity or real-income context.
3. **Validation.** Test whether the composite predicts independent hardship or resilience outcomes and whether it improves on simpler measures.
4. **Uncertainty and geography.** Add ACS margin-of-error handling, band-switch uncertainty, population-weighted alternatives, and a metro/county/PUMA feasibility path.

Insurance and disaster exposure should remain experimental until clean state or local measures are available. Adding a weak proxy would make the map look richer while reducing trust.

10. Conclusion

The revised project should be judged as a measurement prototype, not as an journal-style economics paper. Its current contribution is practical and limited: it shows that a transparent state-level household-pressure layer can be constructed, decomposed, stress-tested, and used to identify where selected public proxies cluster.

The external review's central critique is correct. The index does not yet measure resilience, validate a theory, predict meaningful outcomes, or reveal a new economic mechanism. The revised title and framing therefore narrow the claim. The project remains worth developing, but the next threshold is validation. Until the measure predicts independent household outcomes and improves on simpler benchmarks, it should remain a prototype rather than a public-facing resilience index.

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